

ASHISH KOSANA

Software Engineer, Backend and Full-Stack

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SUMMARY

Backend and full-stack software engineer. I build and ship features end to end, from **Amazon Web Services (AWS)** infrastructure as code and **REST APIs** secured with Cognito to cross-platform **Flutter** and **TypeScript** user interfaces, deployed to production with typed tests that pass in Continuous Integration (CI). Strong in **data structures, algorithms**, and systems fundamentals, and I use **LLMs** to build and ship faster.

SKILLS

- **Languages:** Python, TypeScript, JavaScript, Dart, SQL, Bash, Linux
- **Backend and APIs:** REST, HTTP, JSON API design, FastAPI, SQLAlchemy, pydantic v2, OAuth 2.0, JSON Web Tokens (JWT), Stripe, hexagonal (ports and adapters) architecture, caching, microservices
- **Cloud and Infrastructure (AWS):** Lambda, API Gateway, Cognito, CDK for Infrastructure as Code (IaC), EventBridge, Secrets Manager, IAM, S3
- **Databases:** DynamoDB (NoSQL), PostgreSQL, SQLite (relational database) **Containers:** Docker, docker-compose
- **Frontend:** Next.js, React (server-side rendered Progressive Web App, PWA), Flutter (cross-platform iOS, Android)
- **Testing:** pytest, unit and integration testing, test doubles and fakes, mypy --strict, ruff, evaluation harness (precision/recall/F1)
- **Workflow:** Git, pull request (PR) and code review, Continuous Integration/Continuous Deployment (CI/CD) with GitHub Actions, Agile
- **Core CS:** data structures, algorithms, Object-Oriented Programming (OOP), operating systems, computer networks

EXPERIENCE

Software Engineer (Intern, full-time conversion track) | Crewtron | Tempe, AZ | May 2026 - Present

- Shipped **clock-in/out with geofence auto-tracking and a live crew map** end to end, using a Flutter UI, Python on **AWS Lambda**, and **DynamoDB**. Crew visibility is scoped by role, and presence triggers auto clock-in.
- Designed **REST APIs secured with AWS Cognito** and provisioned the backend as code with **AWS CDK** and scoped IAM policies. Built merchant **review-request flows** over email/SMS.
- Built cross-platform **Flutter features on the Stripe API** (invoicing, quotes, payments) and a **UTC timezone pipeline** that drives a timezone-aware calendar.
- Improved quality with structured QA, **OWASP Top 10 security reviews**, and a consistent Git/PR workflow. Authored an **API integration-test suite reaching ~82% coverage** and integrated LLM-based automation into internal tooling.

PROJECTS

snip | URL-shortener API (pure backend/systems) | FastAPI, SQLAlchemy, SQLite, Docker, no LLM

github.com/Ashishkosana/snip

- Generates short links with a **base62 codec** that has no dependencies and avoids collisions. A **StorePort protocol** lets the SQLite and in-memory backends swap in behind one interface, so the API layer never touches a concrete store.
- Adds a per-IP **token-bucket rate limiter** (429 on exceed) and **TTL link expiry** (410). **28 tests** cover the codec, rate limiting, and expiry, with mypy --strict, Docker, and CI green.

Rythu | Telugu-first farmer weather and crop-advisory PWA (deployed) | Next.js, TypeScript, Python, AWS Lambda, DynamoDB, API Gateway, CDK, Amplify

github.com/Ashishkosana/rythu

- Caches third-party weather forecasts in **DynamoDB with a TTL**, so repeat lookups skip the external weather API. This cuts latency and cost.
- Ships a **server-rendered Telugu PWA** (Next.js on **AWS Amplify**, git-push CI/CD) over a Python **Lambda and API Gateway** backend, with infrastructure as code in **AWS CDK**. A rule-based advisory engine uses GPS and OpenStreetMap geocoding and applies government-verified fertilizer rules that account for nutrient overlap and prevent over-application. **~135 tests** (pytest and Vitest), mypy --strict, CI green.

review-lens | code-review tool that runs multiple review lenses and verifies its own findings | Python, pydantic v2, CLI and GitHub Action

github.com/Ashishkosana/review-lens

- Runs an **LLM across multiple independent review lenses**, then a **verification pass** rechecks each finding before it is reported. This favors precision over recall to cut false positives.
- Includes a **precision/recall/F1 evaluation harness** over a labeled diff dataset, and ships as both a CLI and a GitHub Action. **74 tests**, mypy --strict, CI green.

career-copilot | job-search agent on AWS serverless (deployed live) | Python, AWS CDK, Lambda, DynamoDB, API Gateway, Cognito

github.com/Ashishkosana/career-copilot

- Runs as a **scheduled serverless agent**: a daily **EventBridge** cron fires a Lambda that triages Gmail, scores job matches, and drafts replies with an LLM. Adapters let the Gmail, LLM, and DynamoDB dependencies be swapped for fakes, so tests run the full pipeline without live services (**53 backend tests**).
- Adds a separate **read-API Lambda** behind **API Gateway** with a **Cognito authorizer**. State persists to a **DynamoDB** table partitioned by userId, with keys stored in **Secrets Manager**, all provisioned as code in **AWS CDK**.

EDUCATION

Bachelor of Science (B.S.) in Computer Science | University of Massachusetts Lowell | Graduated Dec 2025

GPA 3.51/4.00 | Dean's List | Transferred from SR Institute of Technology (JNTU), Hyderabad to complete the B.S.